



Modifications of the heat source model in numerical analyses of the metal-cored arc welding process



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<https://doi.org/10.1016/j.energy.2024.131811> 

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Highlights

- A new approach to defining the heat source model is proposed.
- An improvement in the quality of mapping the shape of the molten metal pool was achieved.
- The simplicity of application of the proposed method of heat source model facilitates its use in industrial applications.

Abstract

The paper presents an example of the possibility of modifying a heat source model, which resulted from the differences between the shape of a standard model in the shape of a double ellipsoid and the geometry of a liquid metal pool in the metal-cored arc welding process. Due to the fact that the heat source models implemented in the software do not always allow to reproduce the actual bead geometry, it was necessary to modify these models and signal the possibilities with which the results of the analyses can be improved. The results of modeling with a standard model in the shape of a double ellipsoid as well as a model consisting of two two-ellipsoidal models and modification of heat input by heating the selected elements of the model mesh are presented. The use of the proposed solutions of heat source models, which allow to reproduce the shape of the modeled molten pool, also allows to increase the precision of the calculated maximum values of the heat cycle temperature, reducing the difference by an order of magnitude from 18 to 22% in relation to the values measured for the standard model, to several percent (1.2–3.3%) in the case of combined models.



Keywords

SYSWELD; FEM; Welding; Heat source model; Thermal cycle; Molten metal pool; Metal-cored arc welding

Nomenclature

ε
total radiation emissivity coefficient

T
temperature

T
time

λ
thermal conductivity coefficient,

C
specific heat

ρ
density

P
phase proportion,

i,j
indices of individual phases,

Q
heat source power

$L_{ij}(T)$
transformation latent heat $i \rightarrow j$

A_{ij}

proportion of phase i converted to j per unit time

 q_k

heat flux density

 α_k

heat transfer coefficient

 T_0

ambient temperature

 q_r

heat flux density

 α_r

radiant heat transfer coefficient

 k_B

Stefan-Boltzmann's constant ($5.67 \cdot 10^{-8} \text{Wm}^{-2}\text{K}^{-4}$)

 Q_f, Q_r

volumetric heat flux density in the front and rear part of the model

 Q

the amount of heat introduced by the source

 a_f, a_r, b, c

parameters describing the length, width, and height of the model

 k, l, m

model form factors

 f_f, f_r

parameters describing the distribution of energy in the front and rear parts of the model

x, y, ξ

coordinates describing the position of the model, described by the equation

τ

total welding time

U

arc voltage

I

welding current

v

welding speed

η

coefficient of thermal efficiency of the welding method

1. Introduction

Welding is the most commonly used technique for joining structural metal components in thermal power plants due to its reliable quality [1], simple and fast performance and reasonable price [2]. The welding process introduces a large local amount of heat that melts the wire metal that fills the weld groove and leads to the local melting of the base material [3]. As the melting and resolidification of the metal takes place rapidly, residual stresses and deformations occur within the welded joint and the area close to it [4]. Residual stresses, especially if they are of a tensile nature, reduce the service life of structural components. On the other hand, residual deformations impair the external appearance of the welded structure and cause problems during its final assembly. The removal of residual stresses and deformations is usually carried out by thermal [5] and mechanical procedures [6]. Since during the construction of large thermal power or nuclear power plants, the number of welded joints reaches several tens of thousands, of which a certain number is very critical, it is clear that the costs of removing residual stresses and deformations can represent a significant financial expense.

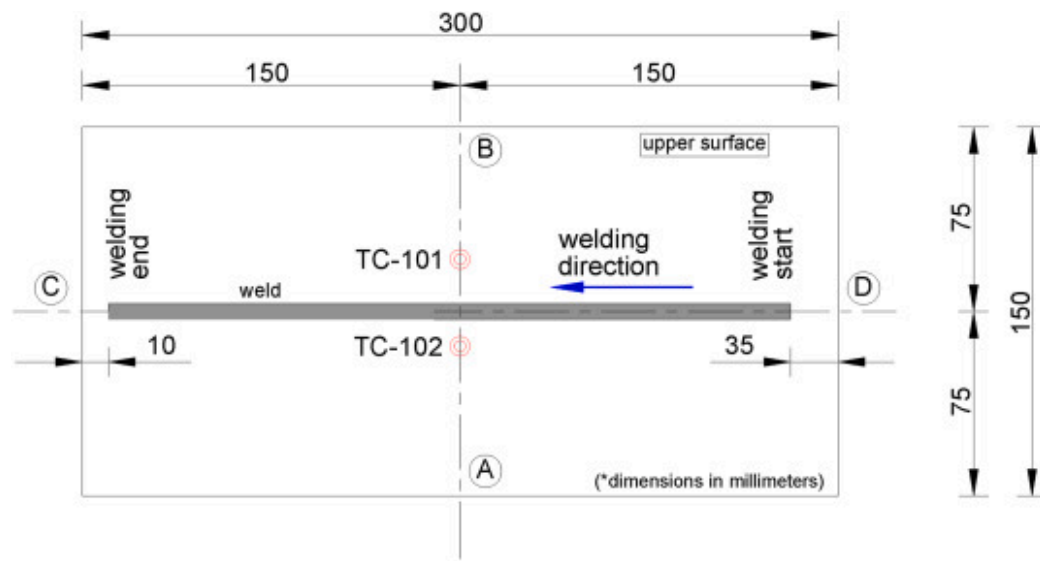
Nowadays, numerical methods are used for the calculation of residual stresses and deformations, which make it possible to reduce the residual stresses and deformations to an acceptable level already during the design process [7]. Typically, numerical welding simulations use a sequential procedure consisting of two independent analyses: thermal and mechanical [8]. Through thermal analysis, the spatio-temporal temperature distribution is obtained from the start of welding until the sample cools down to room temperature [9]. After the thermal analysis is completed, the obtained spatio-temporal temperature distribution is used as a thermal load in the mechanical analysis [10]. In doing so, Jiang et al. [11] carried out a numerical analysis of a wind turbine blade rotor welding and carried out a geometric optimization of the weld between the tower and the bottom flange. Similar to them, Tan et al. [12] investigated the influence of residual stresses on the operating stresses of a welded turbine rotor in a nuclear plant. Feng et al. [13] simulated stress relaxation in a welded turbine rotor under creep and fatigue loadings. In their research, Hemmatzadeh et al. [14] focused on numerical and experimental investigations for the residual stresses of two butt-weld API X46 steel pipes. Forouzan et al. [15] numerically and experimentally studied the distribution of residual stresses on the example of spirally welded pipes. Jiang et al. [16] numerically simulated and experimentally measured residual stresses in their research after charring and after post-weld heat treatment. It can be said that the accuracy of the thermal numerical simulation, and consequently of the mechanical simulation, is largely determined by the fact that the form of the heat source was used in the numerical simulation. This is especially evident in multi-pass welding, as is the case with thick-walled pipes in nuclear power plants. In the literature for the simulation of Metal Inert Gas (MIG) welding, Metal Active Gas (MAG) welding, Tungsten Inert Gas (TIG) welding, Submerged Arc Welding (SAW) and similar arc-welding technologies, the standard double-ellipsoid Goldak's heat source is commonly used [17]. All the mentioned research above carried out some of the listed procedures. The welded sample macrographs which use some of these technologies show a great agreement with Goldak's model, and accordingly, the temperature simulation results are very close to the experimental measurements. On the contrary, Metal-Cored Arc Welding technology (MCAW) [18] due to the specifics of metal transfer and heat conductivity defined by the mixture of inert and active gases causes a macrograph with finger shaped penetration. Such a deformed shape requires the modification of Goldak's heat source, in order to be applicable for thermal field simulation, which is the subject of this research. To the knowledge of these authors, this is the first modification investigation of Goldak's heat source in MCAW welding and can be considered a novelty.

This paper has 6 sections. Section 2 describes the experimental setup and temperature measurement using a thermocouple and thermographic camera. Section 3 is a part that describes the issues of heat transport during welding and achievements in the field of heat source model modification. Section 4 describes the numerical model used in the analyses along with the variants of

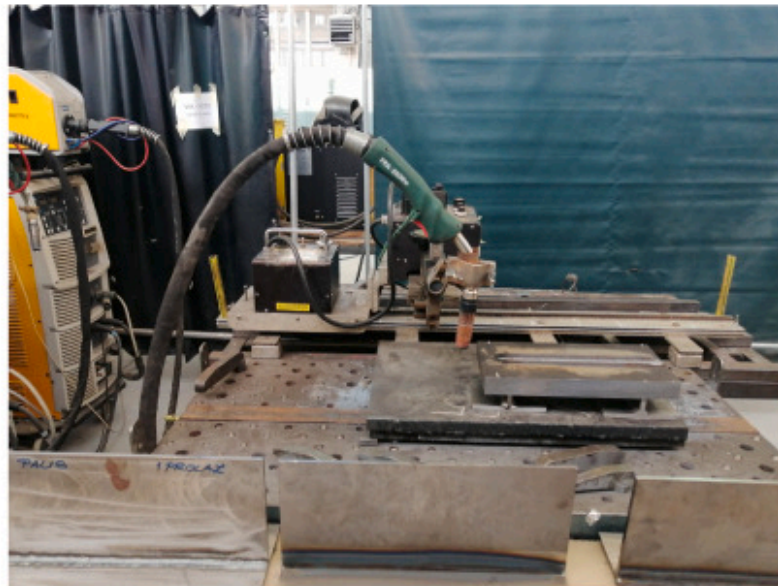
the heat source models used. In Section 5, the results obtained from the numerical simulations were compared with the results of experimental studies. The whole thing is summarized in Section 6.

2. Experimental investigations

A bead-on-plate weld was made on a steel plate made of low-carbon steel 10025-2: S355J2+N using MCAW (ISO 4063:138) welding technology ([Fig. 1](#)). The plate dimensions were 300mm×150mm with a thickness of 30mm. The start of welding was 35mm from the right edge of the plate; the welding end was 10mm from the left edge. The welding parameters were: welding current 320 A, welding voltage 35V, welding speed 57.5cm/min, wire feed speed 14.5m/min and gas flow 20L/min. The shielding gas mixture was composed of 86%Ar/12%/CO₂/2% O₂. The welding power source was Daihen Varstroj WB-P500L and filler wire is classified as EN ISO 17632-A:T 46 2M M4 H5 with a diameter of 1.2mm. As part of the experiment, the temperature measurements were carried out using thermocouples and a thermographic camera. The welding sample was positioned on the welding table on six contact points. This points are physically arranged as ball support made of bearing steel in which contact was made through minimal surface. Regarding to that, there was 2.5mm space between test sample and table so dissipation of heat is minimized during welding.



a)



b)

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Fig. 1. Bead-on-plate welded sample: a) Geometry and dimensions; b) Experimental setup.

The temperature measurement inside of the welded sample was carried out with two K-type (Nickel–Chromium/Nickel-Alumel) thermocouples with a diameter of 1.35 mm, which were placed 7 mm below the upper surface ([Fig. 1](#)) of the sample in pre-drilled holes with a diameter of 2 mm. The thermocouples were designated as TC-101 and TC-102. The distances of thermocouples TC-101 and TC-102 are 13 mm and 10 mm from the weld centerline. The thermocouples were strong pressed to the bottom of the drilled hole and attached to the sample with heat-resistant adhesive tape. The electrical conductors of the thermocouples with a length of 500 mm were connected to the temperature measuring sensors in a tablet Cobra SMARTLink type 12999-99 (PHYWE GmbH, Göttingen, Germany). The temperature measurement in the sample material was carried out with a resolution of 0.1 °C with a relative error of $\pm 3\%$ and a sampling frequency of 10 Hz. The locations of the thermocouples on the welded sample can be seen in [Fig. 1](#).

A thermographic camera VarioCAM HD head 800 (Infratec GmbH, Dresden, Germany), with a standard lens of focal length 1.0/30 mm and an uncooled microbolometric Focal Plane Array (FPA) was used to measure the surface temperature with an IR radiation detector in the format of 1024×768 IR pixels. The thermal resolution of the camera is 20 mK, the measurement accuracy is $\pm 1\%$, and the measuring IR spectrum is from 7.5 μm to 14 μm . The thermographic camera was connected with a GigE-Vision cable to a laptop and computer program IRBIS 3 Plus (Infratec GmbH, Dresden, Germany). To protect the IR lens from damage by dust and melt droplets sprayed from the weld, the camera was placed above the welded sample at a height of 1000 mm and horizontally moved away from the centre of the weld at a distance of 500 mm. Because of this position, a part of the IR beam emitted from the surface of the weld comes into the camera lens, deflected by 30° from the perpendicular axes to the weld surface. Despite this, the field of view (FOV) of the thermographic camera at a distance of 950 mm is 554 mm $\times$ 416 mm, which is sufficient for recording the temperature field on the entire surface of the welded sample with a grid with IR pixel dimensions (IFOV) of 0.54 mm $\times$ 0.54 mm. This camera position is taken into account when reading the measured temperatures in the IRBIS3 Plus program. To measure the temperature with a thermographic camera on the oxidized surface of low-carbon steel 10025-2: S355J2+N in the IRBIS 3 Plus program, the emissivity coefficient of $\varepsilon=0.8$ was set. The ambient temperature during the experiment was 20 °C with an air humidity of 55 %. During the heating, the lens of the thermographic camera was covered with glass and protected against UV radiation from the electric arc. The temperature measurement on the weld surfaces started after

the electric arc was extinguished. The cooling of the weld surface was measured with a thermographic camera for a total duration of 150s with a recording frequency of IR images of 6Hz. The analysis of the surface temperature distribution during the cooling process was carried out in the computer program IRBIS 3 Plus.

Before the welding experiment, the calibration of the surface emissivity coefficient was carried out by heating a calibration sample from 100 to 500°C. During the heating of this sample, the temperature of the surface in the middle and at the edge of the sample was simultaneously measured with type-K thermocouples connected to a Cobra SMARTLink computer and a thermographic camera VarioCAM HD 800 placed above the center. The difference in the temperature value measured by thermocouples and read on the image of the thermal camera of less than 2°C was achieved with an emissivity coefficient of 0.8 ± 0.04 .

3. Basics of heat theory in welding and heat source model development

Numerical simulations have been used for more than three decades to support structural designs and technology development. The reason for this is a significant increase in the complexity of operations in modern production processes and the use of advanced construction materials obtained in complex manufacturing processes, and thus reacting differently a.o. to the thermal cycle of welding processes [19]. The use of numerical simulations also makes it possible to increase the quality of production and the usability of products, while reducing its costs as much as possible. However, for this to be possible, it is necessary to provide a large number of reliable inputs describing the simulated process [20].

In most welding processes, heat is transported by the following heat flow mechanisms: conduction, convection, and radiation [21]. The first of these mechanisms occurs in the case of heat transfer, in which no mass movement occurs. Heat conduction can be seen as the transfer of energy from more to less energetic particles of a substance as a result of the interaction between them. Therefore, if there is a temperature gradient, the energy transport by conduction must then occur in the direction of the region with a lower temperature. Thermal energy is transported by random molecular motion, and this is the dominant mechanism of heat conduction in terms of cooling rates found in welding processes [22]. To describe this phenomenon mathematically, the Fourier thermal conductivity equation is used, which, after being supplemented by taking into account phase transitions by including the latent heat of the transformation and change in the state of matter, takes the form [23]:

$$(\sum_i P_i (\rho C)_i) \frac{\partial T}{\partial t} - \nabla ((\sum_i P_i \lambda_i) \nabla T) + \sum_{i < j} L_{ij}(T) \cdot A_{ij} = Q \quad (1)$$

where:

T - temperature,

t - time,

λ - thermal conductivity coefficient,

C - specific heat,

ρ - density,

P - phase proportion,

ij - indices of individual phases,

Q - heat source power,

$L_{ij}(T)$ - transformation latent heat $i \rightarrow j$,

A_{ij} - proportion of phase i converted to j per unit time.

Heat transport in the welding process is also related to the dissipation of heat to the environment. In this case, two of these mechanisms, i.e. convection and radiation, are taken into account in the simulations of welding processes. Part of the heat introduced into the material during welding is dissipated by convection on the outer parts of the welded elements. The heat flux density is described by Newton's equation [24]:

$$q_k = \alpha_k \cdot (T - T_0) \quad (2)$$

where:

q_k - heat flux density,

α_k - heat transfer coefficient,

T - surface temperature,

T_0 - ambient temperature.

Radiation as a quantity dependent mainly on the shape of the temperature field is determined by the relationship that allows to take into account the occurrence of this method of heat transfer in the range of high temperatures [24]:

$$q_r = \alpha_r \cdot (T - T_0) \quad (3)$$

$$\alpha_r = \varepsilon \cdot k_B \cdot (T + T_0) (T^2 + T_0^2) \quad (4)$$

where:

q_r - heat flux density,

α_r - radiant heat transfer coefficient,

T - surface temperature,

T_0 - ambient temperature,

ε - emissivity of the surface,

k_B - Stefan-Boltzmann constant ($5.67 \cdot 10^{-8} \text{Wm}^{-2}\text{K}^{-4}$).

However, due to the fact that welding processes are usually characterized by high cooling rates, the effect of radiation on the cooling phase, structural and phase changes obtained is relatively small. Thus, in the case of welding, thermal conductivity is the dominant mechanism, and the calculation of temperature fields is usually based on the mathematical definition of the heat source model, represented by the heat flow density in the material. The combination of the heat transfer coefficient and radiant heat transfer coefficient into a single relation allows to solve this problem in a single calculation step by the linear superposition

of heat fluxes [24]:

$$q = (\alpha_k + \alpha_r) \cdot \Delta T \quad (5)$$

where:

α_k - heat transfer coefficient,

α_r - radiant heat transfer coefficient.

In numerical analyses of welding processes, the main and basically only load is the non-stationary distribution of the calculated temperature fields [25]. Therefore, the accuracy of the results of calculations obtained in thermometallurgical and mechanical analyses depends on the accuracy of its description [26]. However, obtaining a weld pool shape that corresponds to the actual one obtained during experiments often requires a large number of input parameters, not only related to process parameters [27]. Such an approach to the issue increases the complexity of software operation, as well as its cost, because it is required to prepare this data in material databases and use much more complex models. Hence, in commercial software, some simplifications of heat source models are used very often [28].

The definition of the heat source includes all process parameters: total thermal energy, the efficiency factor of the welding method, geometry of the molten metal pool, etc. [29]. The most popular three-dimensional models of heat sources currently used in most commercial welding simulation programs are.

- flat source with Gaussian heat distribution,
- heat source in the shape of a double ellipsoid (the so-called Goldak heat source),
- a volumetric source with a Gaussian heat distribution (also called a conical heat source).

Proposed in 1984 by John Goldak, the three-dimensional model of heat source in the shape of a double ellipsoid is still one of the most commonly used predefined models for simulating arc welding processes [30,31]. Despite the aforementioned popularity, its use does not guarantee the correct shape of melting in the welded material [32], as well as the temperature distribution on the cross-section of the welded joint [33].

The analysis of the available literature has shown that there are some attempts to build models that are a combination of standard models as well as completely new ones. These are models with shapes adapted to the selected welding method, but also taking into account additional factors, such as the convection of liquid metal in the weld pool [34]. A significant part of the available studies concerns primarily laser techniques, in which the basic model of the heat source is a volumetric cone model with a Gaussian power distribution. This is mainly due to the high popularity of these methods, as well as the quality requirements for welded elements. The modifications are mainly related to methods in which the shape of the obtained penetration into the welded material differs from the shapes possible to be obtained with the use of the standard Goldak model and the conical model.

In 2006, Wu et al. proposed a modified three-dimensional conical heat source model (TDC) consisting of a Gaussian volume model in the upper part and a conical model in the lower part [35]. Yan Li et al. used the assumptions of this model, replacing the Gaussian-distributed model with the Goldak model. By simulating the plasma welding process of butt joints in X5CrNi18-10 steel sheets, he obtained results with high agreement regarding the conducted experiments. Both of these models have been used mainly in modelling plasma and laser welding using the „keyhole“ technique [36].

In 2014, Chen et al. proposed a heat source model consisting of a double ellipsoid and cylindrical model. On the one hand, the use of a combined heat source model allowed for a much better representation of the penetration line shape in the case of its irregular shape, and on the other hand, it required the determination of a greater number of input parameters [37]. Three years later, T. Flint developed a double ellipsoidal conical heat source model (DEC) as a solution to the problem of narrow-gap TIG welding simulation and the eyelet technique in electron beam welding. This model has also been correctly verified by the results of the tests carried out [38].

F. Farrokhi et al. drew attention to the fact that in the case of hybrid welding with full melting, the shape of the weld may change depending on the process parameters, pointing to the research on laser welding processes conducted by Powell et al., during which a situation in which the lower part of the bead expanded was thoroughly analyzed. The developed combined heat source model in this case consisted of a Goldak model for simulating the arc welding process and a model consisting of two volumetric cone models for simulating laser beam welding. The resulting model of the heat source allowed to get much closer to the actual shape of the beads obtained as a result of welding tests [39].

Also in the case of arc welding methods, models are used that combine two separate models in order to obtain a bead shape corresponding to the real one. An example of this is the use of a model consisting of two ellipsoids arranged horizontally and vertically by D. Podder, in order to reflect the shape of the beads obtained in the submerged arc welding process of structural steel sheets [40].

The cases mentioned above indicate that it is often necessary to combine heat source models in order to obtain the correct shape of the simulated bead. The problem in this case is to record and divide the amount of heat introduced into the individual components of such a combined model. The mathematical notation of the heat distribution into its constituent elements of a combined heat source model is usually complicated. While they can be used in the case of research analyses, they carry a significant risk of creating an overly complex mathematical notation in the case of industrial applications. Being limited to a specific application and having a complex mathematical apparatus can be a real application obstacle.

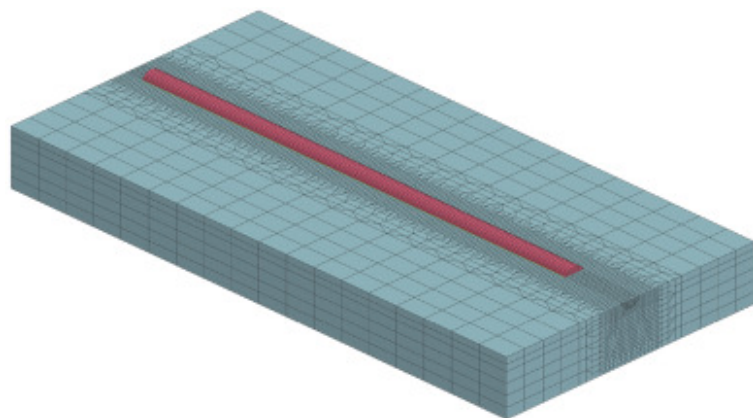
The most important part of the input data in welding process analyses is the variable nature of the moving heat source "load". To correctly define the mathematical notation of the heat source model, it is necessary to provide as many input parameters as possible that affect the way the material is heated. Validation of the model, in turn, is carried out on the basis of the available possibilities of conducting real experimental trials. When determining the heat source model and its input parameters, the results are verified by Ref. [41].

- comparison of the shape and dimensions of the calculated molten metal area and HAZ on the basis of macroscopic examinations,
- comparison of recorded and calculated heat cycles,
- comparison of the shape and dimensions of the liquid metal pool surface with the recorded thermographic images.

Since the shape of the molten metal pool and heat distribution in the junction determine the resulting phase distributions, and consequently also the stress and strain distributions, there is still a gap in the area of knowledge regarding the correct representation of the heat source model. Its supplementation forces the search of new solutions, which, due to the application-based nature of the analyses conducted, cannot be burdened with an excessively extensive mathematical apparatus.

4. Numerical model and assumptions

As part of the research, numerical analyses of the MCAW welding process were carried out in order to determine the method of preparing the heat source model, and as it turned out in the course of the study, the necessary modifications of the heat input method and development of a methodology for conducting the construction, calibration and validation of the heat source model in order to obtain correct results. For this purpose, a three-dimensional discrete model consisting of 111,798 solid elements and 110,070 nodes was built, [Fig. 2](#).

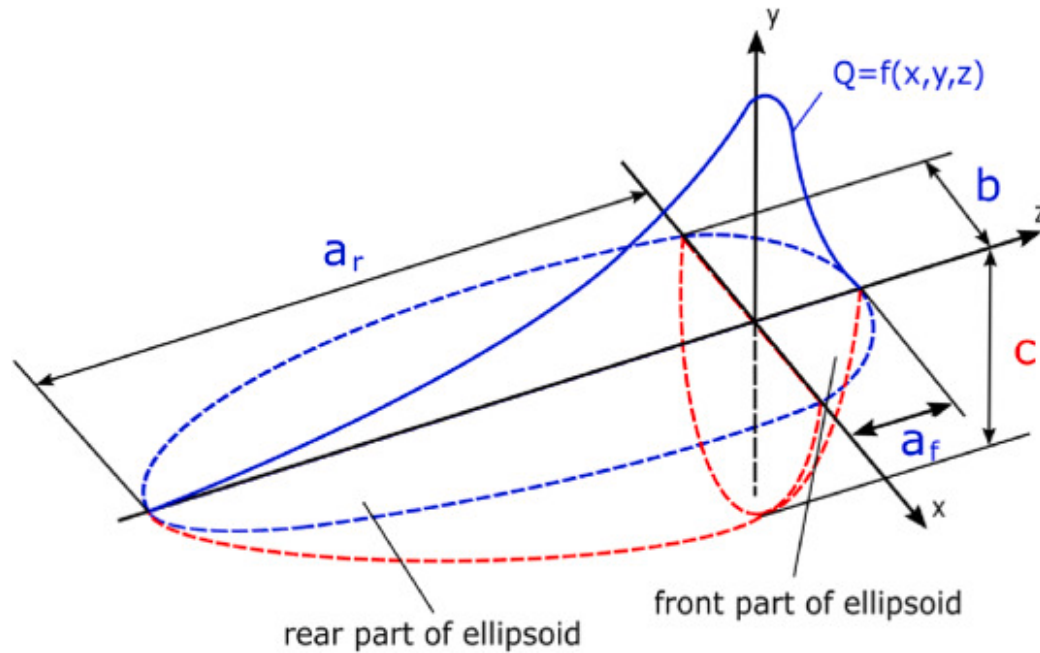


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Fig. 2. View of three-dimensional numerical model.

In order to increase the precision of the calculations, the mesh of the model has been compacted in the area of bead and heat-affected zone. The applied boundary conditions simulating the clamping of the sample corresponded to its free placement on the welding table. The boundary condition describing the heat dissipation to the environment was defined by radiation and convection heat dissipation to an ambient temperature of 20°C on all the external surfaces of the model. For these surfaces, the temperature-independent coefficient of convective heat transfer $\alpha=25 \text{ Wm}^{-2}\text{K}^{-1}$ was assumed, while the surface emissivity value of 0.8 was determined by experimental measurement. The thermal analysis of the process was carried out as a continuous 'transient' analysis, using a moving, three-dimensional model of heat source in the shape of a double ellipsoid, also called the Goldak model, [Fig. 3](#). This model is the most commonly used heat source model in the simulation of arc welding processes [30].



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Fig. 3. Model of heat source in shape of double ellipsoid.

The mathematical description of the Goldak model consists of two equations that describe the front and rear parts of the ellipsoids independently [30]:

for the front of the model:

$$Q_f(x, y, z) = \frac{6\sqrt{3}f_f Q}{a_f b c \pi \sqrt{\pi}} \exp\left(\frac{-kx^2}{c^2}\right) \exp\left(\frac{-ly^2}{b^2}\right) \exp\left(\frac{-mz^2}{a_f^2}\right) \quad (6)$$

for the rear of the model:

$$Q_r(x, y, z) = \frac{6\sqrt{3}f_r Q}{a_r b c \pi \sqrt{\pi}} \exp\left(\frac{-kx^2}{c^2}\right) \exp\left(\frac{-ly^2}{b^2}\right) \exp\left(\frac{-m \cdot (z - v(\tau - t))^2}{a_r^2}\right) \quad (7)$$

where:

Q_f, Q_r - volumetric heat flux density in the front and rear part of the model.

Q - the amount of heat introduced by the source

a_f, a_r, b, c - parameters describing the length, width, and height of the model

k, l, m - model form factors

f_f, f_r - parameters describing the distribution of energy in the front and rear parts of the model,

$$f_f = \frac{2}{\left(1 + \frac{a_r}{a_f}\right)}, f_r = \frac{2}{\left(1 + \frac{a_f}{a_r}\right)}, f_f + f_r = 2 \quad (8)$$

x, y, ξ - coordinates describing the position of the model, described by the equation:

$$\xi = z - v(\tau - t) \quad (9)$$

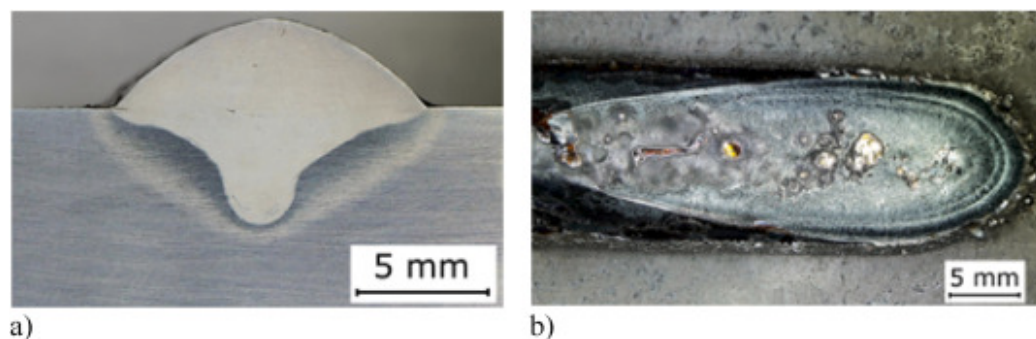
τ - total welding time.

Goldak's source can be used in its standard and modified form. In its standard form, the coefficients k, l, m are constant values and amount to 3. However, the first attempts of analysis showed that the model in this form could not be used as a universal heat source model for all welding methods. In his subsequent works, Goldak included information about the necessity and possibility of its modification [26,31].

According to the mathematical description, the equations describe the front and back parts of this heat source model separately, so four dimensions are needed to determine its geometry: width, depth, and length of the front and back of the ellipsoid. The width and depth of the bead penetration and length of the molten metal pool are therefore the inputs necessary for the initial definition of this heat source model. According to Goldak's assumptions, the length of the front part of the ellipsoid is equal to half the width of the bead, and the length of the back part is twice the length of the bead [42]. In later papers, a modification of this assumption can be found, comparing the length of the front part to the depth of the penetration and defining the length of the rear part as four times the length of the front part [27,29].

Since the VisualWeld (Sysweld) software used has predefined profiles for selecting the heat source model, a MAG method profile was used. This model automatically divides the entered length of the molten metal pool in a ratio of 1/3 to 2/3, and do not

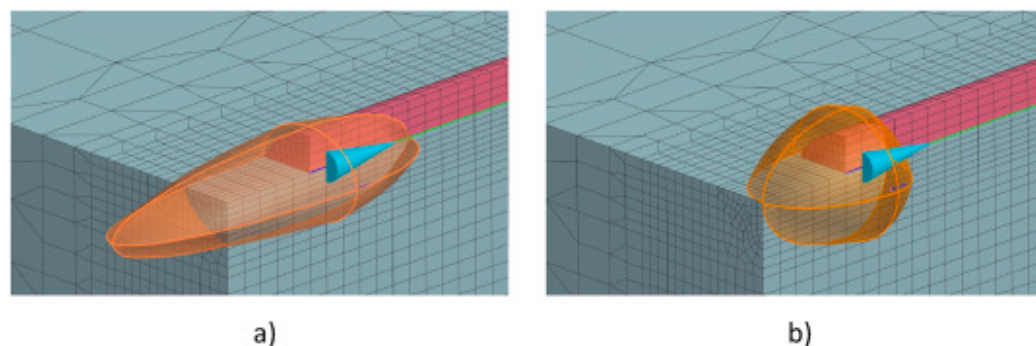
depend on the width of the bead. Calculations of the molten metal pool dimensions were carried out without changing this proportion, defining its dimensions on the basis of the measurement results obtained from real welding tests, Fig. 4, Fig. 5a.



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Fig. 4. View of a) macrostructure and b) surface of molten metal pool of MCAW weld bead used in research



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Fig. 5. View of three-dimensional model of heat source in shape of double ellipsoid: a) model with dimensions from experiment, b) model with modified dimensions (M1 model).

Due to the differences in the calculated and actual dimensions of the liquid metal pool, calculations were also carried out, in which the individual dimensions of the standard heat source model were selected in such a way to obtain the closest possible to

the actual values of the calculated molten metal pool dimensions (M1 model), [Fig. 5b](#).

In the case of continuous analysis, the energy introduced into the welded parts is the energy generated by the moving heat source and absorbed by the welded component with a certain efficiency. The amount of heat introduced into the welded joint is given by the relation:

$$Q = \frac{U \cdot I}{v} \cdot \eta \quad (10)$$

where:

Q - heat source power,

U - arc voltage,

I - welding current,

v - welding speed,

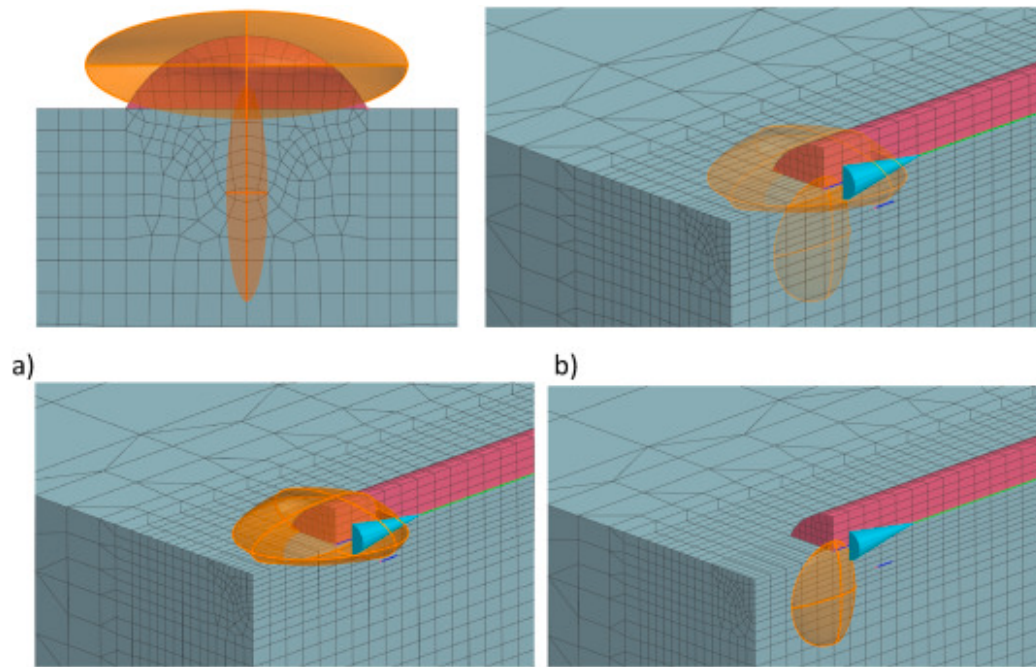
η - coefficient of thermal efficiency of the welding method.

In the thermal numerical simulation, the coefficient of thermal efficiency in the MCAW process was based on a standardized value of $\eta=0.8$. For each of the model variants analyzed, the amount of heat input into the weldment was $Q=990\text{J/mm}$.

Because the shape of the heat flux during the MCAW, it is significantly different geometrically from the Goldak model, and the heat distributions calculated using the standard Goldak model do not correspond to the macrostructure image of the analyzed bead. Therefore, attempts were made to modify the method of heat input so that it could be reproduced as well as possible. There is no ready-made solution in the scientific literature suitable for simulating the MCAW welding process, but there are examples of creating a heat source model both by changing the equations describing its structure and operation, and based on its assembly from several standard models [[43](#)].

The latter, the mentioned variant, was the second of the models used in the presented analyses, marked as the M2 model. In order to reproduce the characteristic „finger shape“ penetration in the MCAW method, the movable model of the heat source was constructed from two standard Goldak models, positioned: horizontally in the upper part of the model and vertically to the lower

part of the model, Fig. 6. The dimensions of both models as well as the breakdown of the total amount of heat introduced into the model were determined on the basis of iterative attempts to calibrate the heat source model until the correct result was obtained.



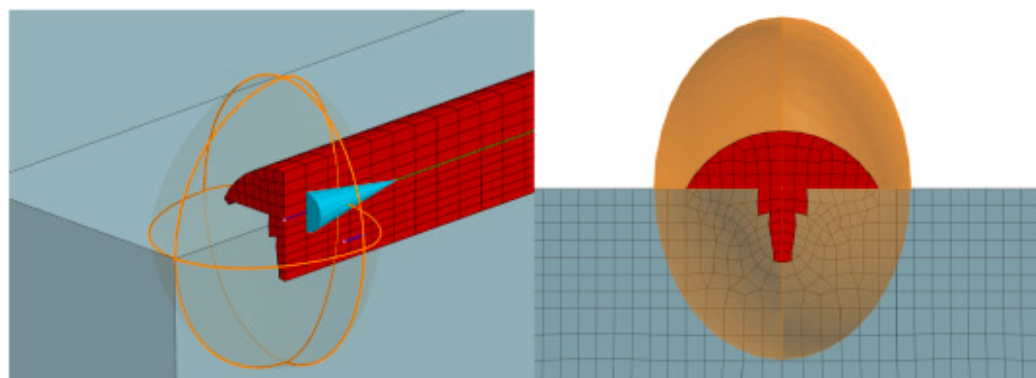
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Fig. 6. View of combined three-dimensional model of heat source, consisting of two Goldak models: a) upper double ellipsoid and b) lower double ellipsoid – M2 model.

A third variant of a heat source model under consideration was the use of the Author's original solution [41], which consisted of combining a standard model in the shape of a double ellipsoid with the possibility of loading selected elements of the model's mesh (M3 model). In this case, the heat source model was defined in such a way that its transverse dimensions completely covered the indicated area of the loaded elements, while the shape of the heat-affected zone was determined on the basis of the weld shape obtained from the macroscopic sample, Fig. 4, Fig. 7. This solution allowed for the heating of individual, selected areas

of the model mesh and obtaining a penetration shape that cannot be achieved with a standard model of heat source in the shape of a double ellipsoid. Its correct application, like in previous models, requires several iterations in the model calibration process.



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Fig. 7. View of modified model of heat source consisting of Goldak source and defined area of selected elements heated by this source – M3 model.

A third variant of the considered heat source model was the use of a proprietary solution, which consisted in combining a standard model in the shape of a double ellipsoid with the possibility of loading selected elements of the model mesh (model M3). This model allows the user to freely shape the area of the model into which heat is introduced. Based on the macrostructure image, the user can adjust the indicated area of the elements that are heated with a standard double ellipsoid model, as well as modify this shape if it still needs to be tuned, Fig. 4, Fig. 7. In such a case, the heat source model is defined in such a way that its transverse dimensions cover the area of elements marked as loaded. Its length is shortened to precisely heat only these elements. Undoubtedly, a big advantage of this solution is that calibration is much easier and less time-consuming compared to the solution used in the M2 model. The changes are mainly in the area of the loaded elements, where in the case of the M2 model it is necessary to calibrate the component models separately, taking into account the fact that the resulting temperature distribution is a value coupled with the calibration results of both models. In the end, the results obtained from the use of the proposed technique are not achievable in the case of using a standard model of the heat source (model M1).

The models prepared in this way were used in a comparative analysis with the results of an actual welding process. The individual dimensions and parameters of the models used are presented in [Table 1](#).

Table 1. Parameters of applied heat source models used in analysis of MCAW welding process of steel samples EN 10025-2:S355J2+N with cored wire EN ISO 17632-A:T 46 2M M4 H5.

Model designation ^a	a , mm	a_f , mm	a_r , mm	b , mm	c , mm	E , J/mm	η , -
M1	15	5	10	6,5	6,0	1168	0,8
M2 ^b	15 (8)	5 (2.667)	10 (5.333)	7.5 (2.0)	2.5 (5.0)	630 (400)	0.8 (0.8)
M3	10	3.333	6.667	7.5	10	1168	0,8

a , b , c – dimensions of the Goldak model (a_f/a_r – length of the front/back part of the ellipsoid), E – linear energy of the welding process; η – thermal efficiency coefficient of welding method.

a

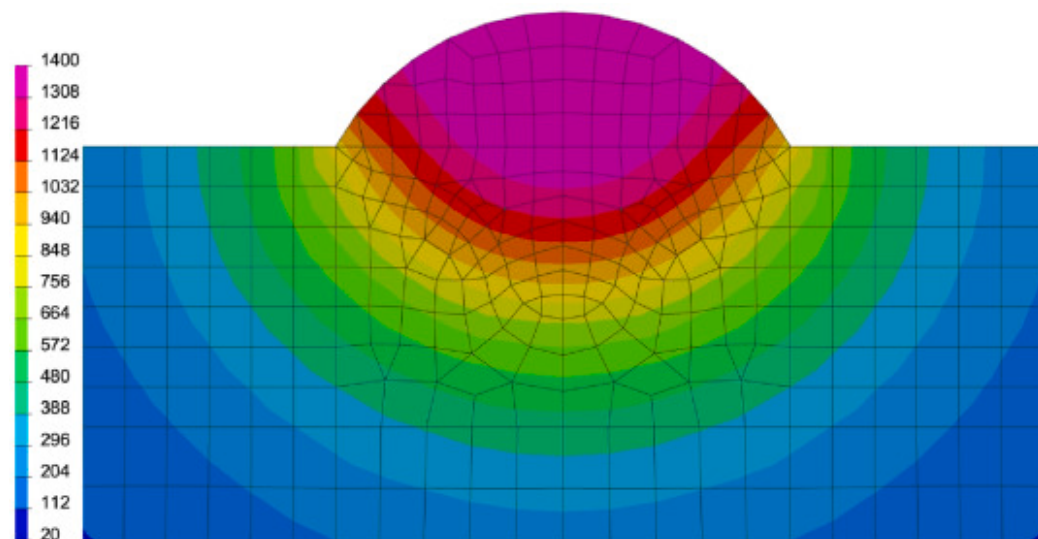
M1 – standard Goldak model, M2 – heat source model combined of two Goldak models, M3 – the modified model of the heat source consisting of the Goldak source and defined area of mesh elements heated by this source;.

b

For a combined model, the values for the lower part of the model are given in parentheses.

5. Results and discussion

As mentioned above, the numerical analyses carried out using a standard Goldak model, with dimensions corresponding to the measured dimensions of a liquid metal pool, did not allow to obtain correct results. On the cross-section of the simulated bead, a significant limitation of the area corresponding to the pool of liquid metal is visible, [Fig. 8](#).

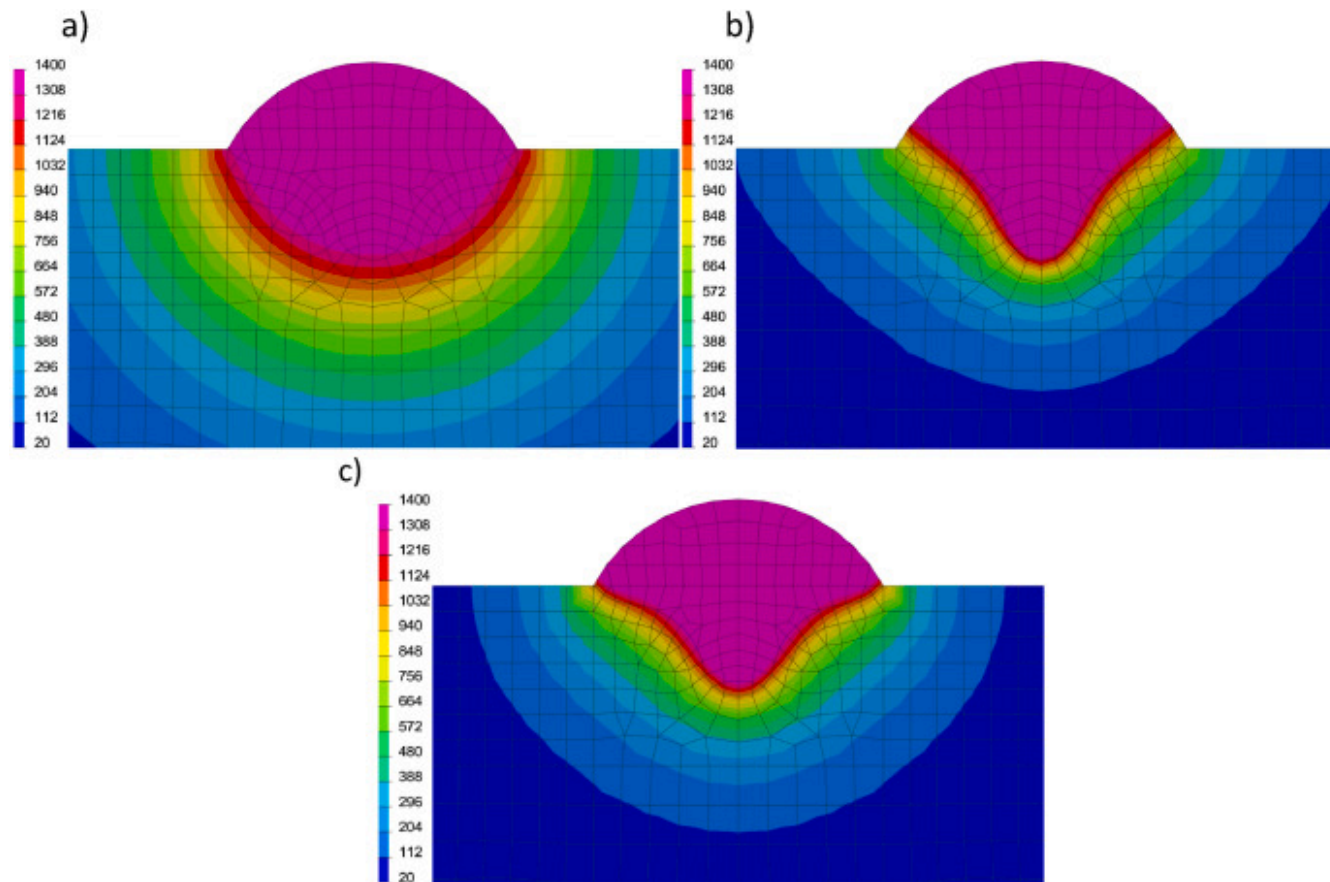


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Fig. 8. View of temperature field distribution in the case of using Goldak model with dimensions corresponding to actual, measured dimensions of molten metal pool.

This result therefore indicates that the use of a double ellipsoid model in an unmodified form cannot be universally used in a wide range of simulations for different welding processes. Of course, it is possible to obtain the correct results of the analysis, but this entails the need to enter values of the model dimensions that differ from the values obtained as a result of the conducted experiments. Such an approach may be incomprehensible to a novice user of the software and additionally prolongs the model calibration stage due to the need to search for these values. The solution to this problem is to use a modified double ellipsoid source model by changing the form of the mentioned factors „ k , l , m “ – unfortunately also by trial and error in multiple iterations. Moreover, in both of these cases, it will still not be possible to obtain the „finger shape“ of the penetration characteristic of MCAW, despite obtaining the correct values of bead width and penetration depth, Fig. 9a.



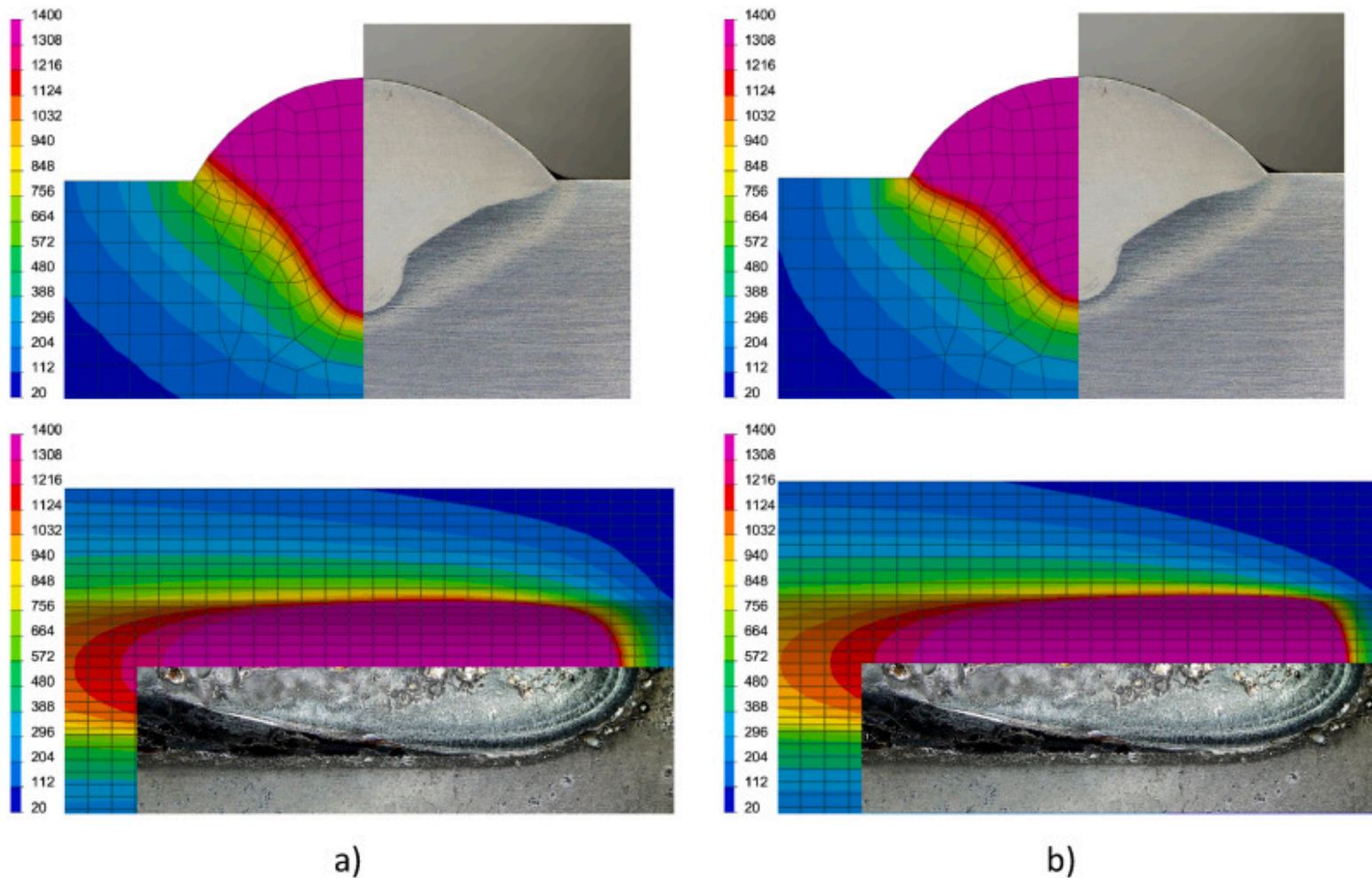
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Fig. 9. Comparison of temperature field distributions of bead modeled with: a) standard Goldak model (M1 model), b) combined three-dimensional model of two Goldak models (M2 model) and c) modified model of heat source consisting of Goldak source and defined area of mesh elements heated by this source (M3 model).

The use of a composite three-dimensional model of heat source, consisting of two Goldak models (M2 model) significantly improved the representation quality of the actual shape regarding the modeled molten metal pool, Fig. 9, Fig. 10a. Both the shape and dimensions of the bead correspond to the values measured in the macrostructure image. The insufficient remelting in the corners of the deposit visible in Fig. 9b is caused by the fact that assembling the model from two sources causes the "upper"

source to heat up the material with a small distance in relation to the "lower" one, melting these areas. However, despite the high accuracy of the results, this model significantly affects the complexity of the heat model preparation and calibration stage. This is due to the necessity of correctly dividing the amount of heat introduced into the two models, as well as the correct selection of dimensions and the position of the component model. This requires many repetitions of calculations, and often changing the parameters of one model affects the total heat distribution in the modeled bead. Nevertheless, as mentioned above, this technique allows for reliable and precise calculation of the shape and dimensions of the modeled bead/molten metal pool.



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Fig. 10. Comparison of temperature field distributions with view of macrostructure and surface of molten metal pool modeled with: a) combined three-dimensional model of two Goldak models (M2 model) and b) modified model of heat source consisting of Goldak source and defined area of mesh elements heated by this source (M3 model).

Compared to the proposed technique of assembling a heat source model from two models in order to obtain an appropriate molten metal pool shape, a simplification consisting of using one heat source model and operating the shape of the model mesh area heated by this source (M3 model) also allows to obtain a high consistency of the simulated bead shape, [Fig. 9](#), [Fig. 10b](#). This technique also significantly speeds up the preparation and calibration of the computational model. This is because the user does not have to define two models, and in addition, there is no mentioned influence of one model on the other. A certain inconvenience of this solution is the "flattening" of the calculated, front part of the molten metal pool ([Fig. 10b](#)), resulting from the fact that by heating the indicated elements of the mesh with the Goldak source is much shorter than in the case of the other variants discussed. The reason for this is that only the selected elements can be heated.

Validation of the applied models on the basis of a dimension comparison measured on a real sample with calculated values, showed a relatively high accuracy of the proposed solutions – in particular, the M3 model due to a significant reduction in the model calibration time as well as the ease of model preparation. The differences in the calculated values of molten metal pool width were up to approx. 3.5%, penetration depth approx. 5.1% and length approx. 6.6%, [Table 2](#).

Table 2. Influence of applied variant of heat source model on geometrical dimensions of molten metal pool in analyses of MCAW welding process of steel sample EN 10025-2:S355J2+N with cored wire EN ISO 17632-A:T 46 2M M4 H5, [Fig. 9](#), [Fig. 10](#).

Model designation	w, mm	d, mm	l, mm	W, mm	D, mm	L, mm
M1	11.555	4.095	36.031	11.710	4.313	33.8
M2	11.305	4.313	35.490			
M3	11.305	4.093	32.787			

w/d/l – width/depth/length of calculated molten metal pool.

W/D/L – width/depth/length of real molten metal pool.

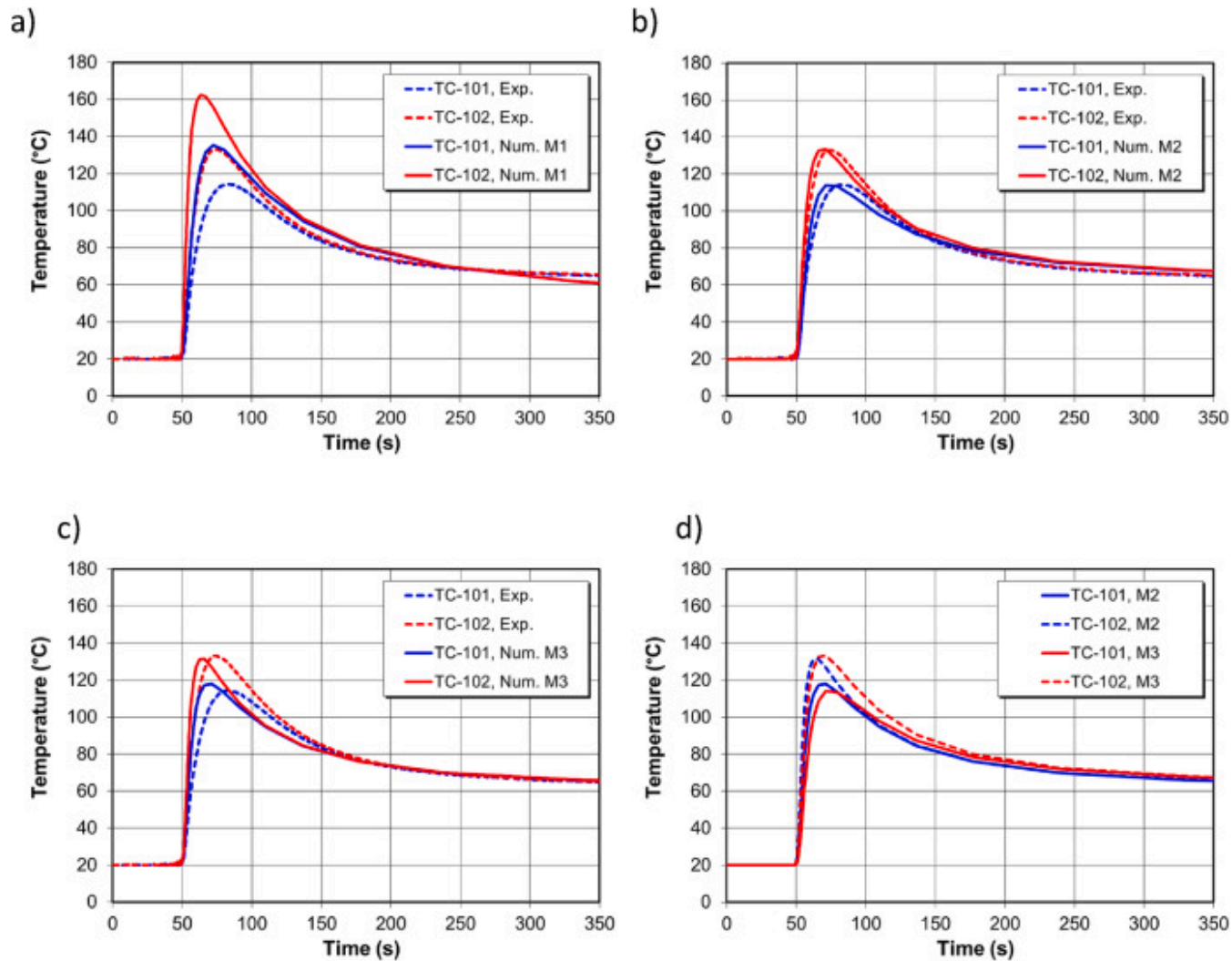
The difference in the time needed to calibrate the M2 and M3 models is due to the fact that in the case of the M2 model, two models in the shape of a double ellipsoid are calibrated, whose interaction with each other at the same time on the welded material results in obtaining the resultant temperature field. However, it is not possible to determine how much this procedure will be shortened, as it depends on the specific case, the complexity of the geometry and the parameters of the welding process.

Changing the parameters of only one of the models (in case of M2 model) also affects the temperature distribution in the environment of the other model. Therefore, it requires many iterations and mutual adjustment of the model to finally achieve this distribution. In the M3 model, the heat source model heats up selected mesh elements. It is up to the user to decide which and how many of them to define, obtaining a given temperature distribution. Calibration is much faster, due to the fact that it does not have to repeatedly change the proportion of the total amount of heat distribution between two models (it is not always the case that it is divided in a 50/50 ratio or the total value is the same as in the case of one model – this is due to the fact that the heat source model is assembled from two separate models). The solution proposed in the case of M3 is therefore much simpler.

The results of the numerical analyses presented above with the proposed variants of heat source modeling clearly indicate the necessity to have an image of the macrostructure of the simulated bead at the stage of preparing the calculation model and calibration of the heat source model. This allows for the correct reproduction of its shape and dimensions. Therefore, the numerical analyses of the welding process should not be treated as a "black box" which, after providing typical parameters of the welding process, will calculate, for example, the depth of penetration and, what is more, indicate the correct shape of the penetration line. This is impossible because the modeling of welding processes overlooks many important factors that affect the shape of the penetration line and final bead dimensions. These are parameters such as the type and flow rate of the shielding gas, the type and diameter of the wire, the method of metal transfer in the arc, etc. Therefore, a properly defined and calibrated model of the heat source and the obtained, correct temperature field distributions should in a sense be treated as an "input data" for the further analyses of metallurgical phase distributions, stresses or deformations.

However, each defined and calibrated model of a heat source should be validated to confirm the correctness of the assumptions made for its construction and the calibration process carried out. Since in the numerical simulations of welding processes, the temperature distribution in the model is mainly determined by thermal conductivity, the easiest way to perform validation is on the measured and calculated temperature values.

The comparison of the calculated thermal cycles of the MCAW welding process for the three analyzed variants of the heat source model construction with the values measured during the experiment by thermocouples TC-101 and TC-102 (Fig. 1a) showed that the thermal cycles calculated using the standard Goldak model (M1 model) are characterized by a much higher maximum value of calculated thermal cycle temperature, Fig. 11a. The highest agreement of calculated and recorded thermal cycles was observed in the case of a combined three-dimensional heat source model, consisting of two Goldak models (M2 model), Fig. 11b. In the case of using a modified model of a heat source consisting of a Goldak source and a defined area of elements heated by this source (M3 model), a steeper course of heating curves and a shorter cycle time are visible, although the waveforms of the cooling curves are also consistent with the measured curves, Fig. 11c. This is due to the aforementioned "flattening" of the front part of the calculated molten metal pool. Taking into account the ease of preparation of this model and the significant acceleration of the calibration stage of the heat source model, which has been mentioned many times before, this model may undoubtedly be easier to use in everyday industrial practice, where small differences in relation to a more labor-intensive model (Fig. 9b) are not as important as in the case of scientific applications.



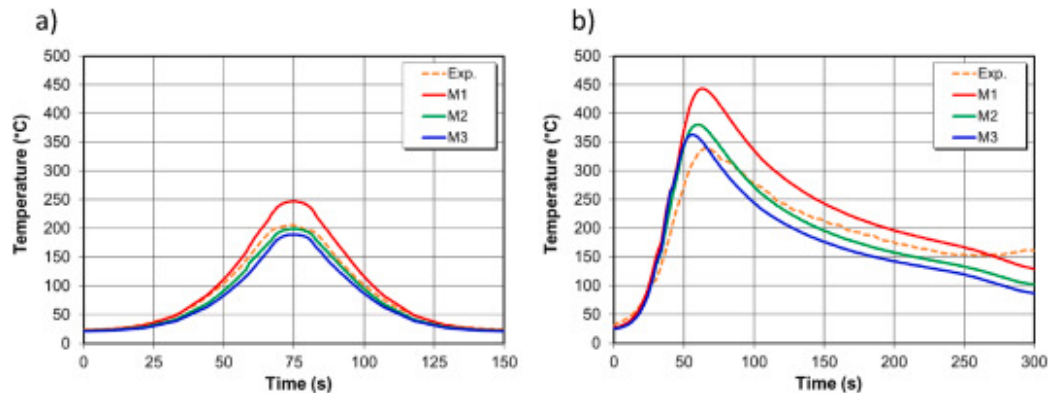
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Fig. 11. Comparison of recorded and calculated thermal cycles in MCAW welding process for application variant: a) standard Goldak model (M1 model), b) combined three-dimensional model of two Goldak models (M2 model) and c) modified model of heat source consisting of Goldak source and defined area of mesh elements heated by this source (M3 model) and d) comparison of M2 and M3 models.

The use of a standard model of a heat source in the shape of a double ellipsoid resulted in obtaining from 18.5 to 22.2% higher maximum temperature value of the calculated heat cycle in relation to the measured values. The use of the M2 and M3 models resulted in a reduction of the difference in the maximum value of the cycle temperature to 1.2–3.3% in the case of the combined model (M2 model) and practically complete agreement (0.3%) in the case of the model with the defined area of the loaded mesh elements (M3 model).

In the process of validation for the used models of heat sources, it is also possible to use the registered thermographic images, Fig. 12. However, this variant brings with it the disadvantage that only one temperature distribution on the surface of the welded material is subject to validation. A comparison of recorded and calculated temperature profiles in lines A-B and C-D (Fig. 1a) on the surface of the welded element 5s after the end of the welding process also showed a high agreement of the simulation results with the measurement values, Fig. 10. As in the case of the recorded thermal cycles, there are characteristically higher maximum profile temperatures for the use of the standard Goldak model (M1 model) and a much better representation of these profiles when using the other two proposed models (M2 and M3 model). The differences in the case of the standard Goldak model range from 20 to 30.4%, while the use of combined models reduces this difference to 1.4–3.1% in the case of the combined model (M2 model) and 6.9–8.0% in the case of the defined area of the loaded mesh elements (M3 model).



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Fig. 12. Recorded thermographic image of bead at 5s after welding and curves of temperature profiles: a) along line A-B on Fig. 1; b) along line C-D on Fig. 1.

As mentioned earlier, a properly defined and calibrated heat source model allows to provide greater precision in the analysis of the effects of the welding process thermal cycle on the properties of welded joints. Of course, one should not forget about the exact reproduction of the shape of the modeled bead, which will undoubtedly be important in the calculation of the values and distributions of the calculated stresses and deformations of welded joints.

The tests were carried out on a sample of S355J2+N structural steel, but such simulations can be easily carried out on materials for applications in the energy industry, provided that a material database corresponding to this material is used (which is available). In this case, it was rather about presenting the issue of modification of the heat source model for the welding method popular in the production of elements and structures for the power industry. As far as wider applications are concerned, the proposed methods of modeling the heat source do not exclude their use for various types of joints (butt, T-joint, lap joint, etc.) as well as welds (butt, T-weld).

6. Conclusions

On the basis of the presented research, it was found that:

- The use of standard models of heat sources does not always ensure the correct dimensions and shape of the molten metal pool are obtained. Even simple bead geometries often require the modification of the heat source mathematical description of the model used in order to correctly represent the dimensions of the molten metal pool.
- The use of combined heat source models, consisting of several standard models, allows to improve the quality of the actual shape reproduction of the bead, but also complicates and prolongs the calibration process due to the need to divide the amount of heat input into component models. However, it is difficult to determine the benefits of reducing the calibration time of the model, due to the fact that the number of interventions in the calibration process depends on many factors such as the shape of the bead, welding process parameters, etc. Usually, however, it is at least a twofold reduction in the time needed to obtain the correct geometry of the calculated liquid metal pool.
- The proposed solution of the heat input method with a defined area of the mesh elements loaded with the use of a standard heat source model allows to significantly accelerate the stage of defining and calibrating the

created heat source model and ensuring the possibility of obtaining any shape of the molten metal pool on the cross-section, while at the same time a high degree of compliance with its actual shape obtained as a result of experimental tests.

- The use of the proposed solutions of heat source models, which allow to reproduce the shape of the modeled molten pool, also allows to increase the precision of the calculated maximum values of the heat cycle temperature, reducing the difference by an order of magnitude (from 18 to 22% in relation to the values measured for the standard model, to several percent (1.2–3.3%) in the case of combined models).
- Comparison of heat distributions determined in the transverse and longitudinal axis of the molten metal pool also showed that the use of the combined heat source models allows to obtain a much higher accuracy of the representation of temperature distributions calculated and recorded with a thermal imaging camera. For the standard Goldak model, the values of max. profile temperature were 20–30% higher than measured, and in the case of combined models, the difference was 1.4–3.1% less in the case a two-source model and 6.9–8.0% less in the case of the model with a defined area of loaded elements.
- Having an image of the macrostructure of the bead is necessary in order to correctly select the heat source model, especially in the case of welding methods in which the shape of the bead takes on complex shapes.
- The proposed method can be successfully used for various types of welded joints, as well as materials used in the construction of equipment for the energy sector, but it is necessary to use properly prepared material databases corresponding to these materials.
- It is also necessary to carry out further studies in a wider range of welding parameters and to determine the influence of the applied heat source model on the calculated stresses and welded joints distortions distributions.

The next goal of the research could be focused on the mathematical modification of the MCAW heat source model and its application to more complex welded structures.

CRediT authorship contribution statement

Tomasz Kik: Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis, Conceptualization.
Ivica Garašić: Writing – original draft, Resources, Investigation, Data curation. **Mato Perić:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Darko Landek:** Writing – original draft, Visualization, Resources, Funding acquisition, Data curation. **Maja Jurica:** Visualization, Project administration, Data curation. **Zdenko Tonković:** Writing – review & editing, Resources, Funding acquisition, Data curation.

Declaration of competing Interest

The authors declare no conflict of interest.

Aknowledgements

This work has been supported by the European Union through the European Regional Development Fund, Operational Programme “Competitiveness and Cohesion” 2014–2020 of the Republic of Croatia, project “Improvement of high-efficiency welding technology” (ImProWE, KK.01.1.July 1, 0076).

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Data availability

No data was used for the research described in the article.

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